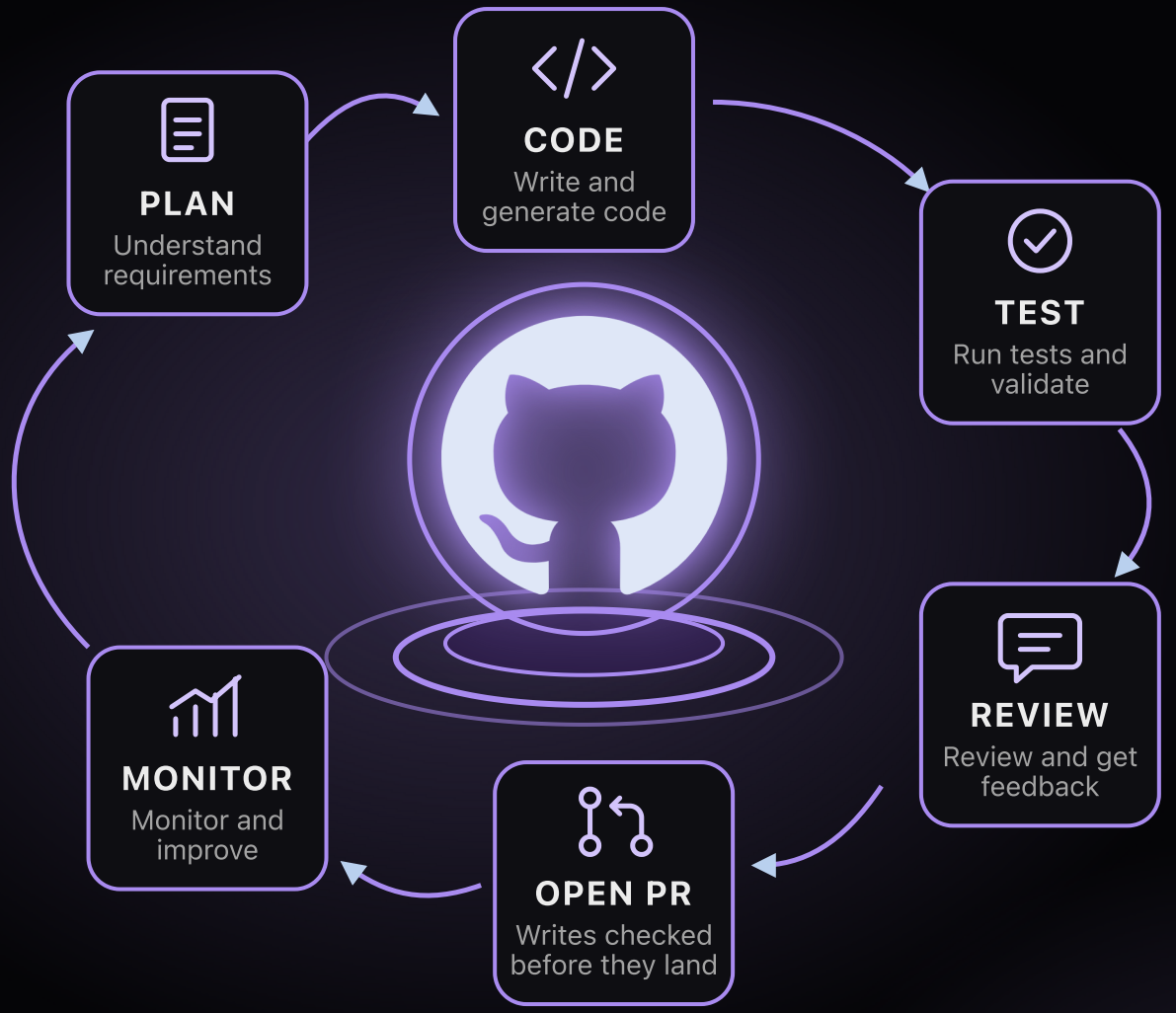
GitHub

GitHub Agentic Workflows

FROM AI THAT SUGGESTS TO AI THAT DOES



01 / THE PROBLEM

Outdated documentation already costs developers time

56.5%

of 12,374 developers say **outdated information** makes it hard to get the context they need.

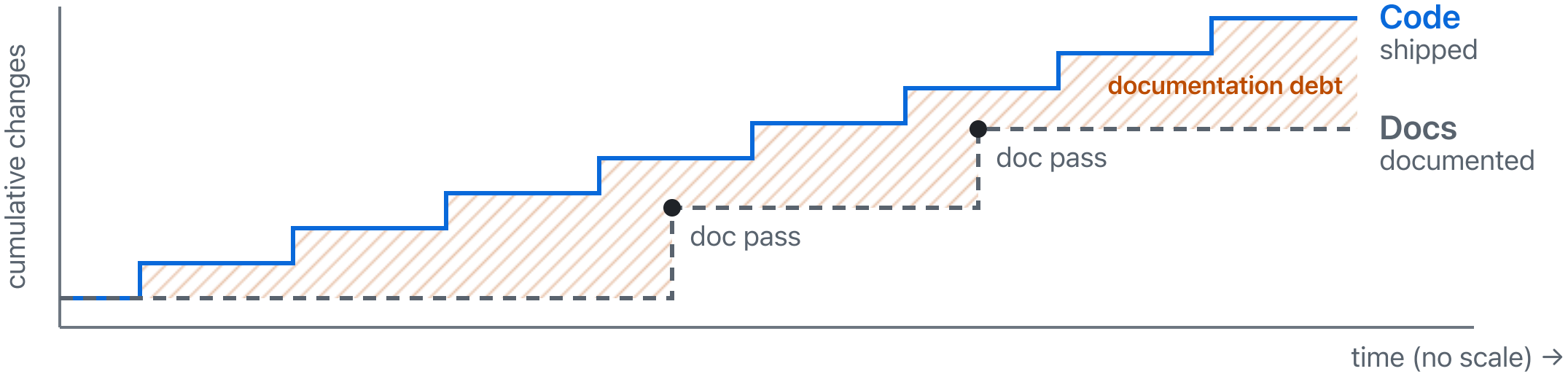
My earlier attempt

It tackled part of the problem.
TODO: the tool, what it solved, and the gap it left.

Industry survey, not our repositories · our own drift is not yet measured.

Documentation debt accrues between doc passes

ILLUSTRATIVE · NO DATA



> Every shipped change adds to it; each doc pass pays only some back.

Fig. 1 · Illustrative · no data · extends Cunningham's (1992) technical-debt metaphor to documentation.

03 / THE ASK

The ask is four weeks, one repository, and docs only

Four weeks. One repository. Documentation only. Every change arrives as a draft pull request, and a person decides whether it merges.

> Everything after this slide is evidence for or against that ask.

TODO · repository, owner, budget and week-four thresholds are still blank.

This deck's proposal · the blanks are what I need from you · the bounds are on the decision slide.

04 / THE SOLUTION

Agents shift the human role from execution to review

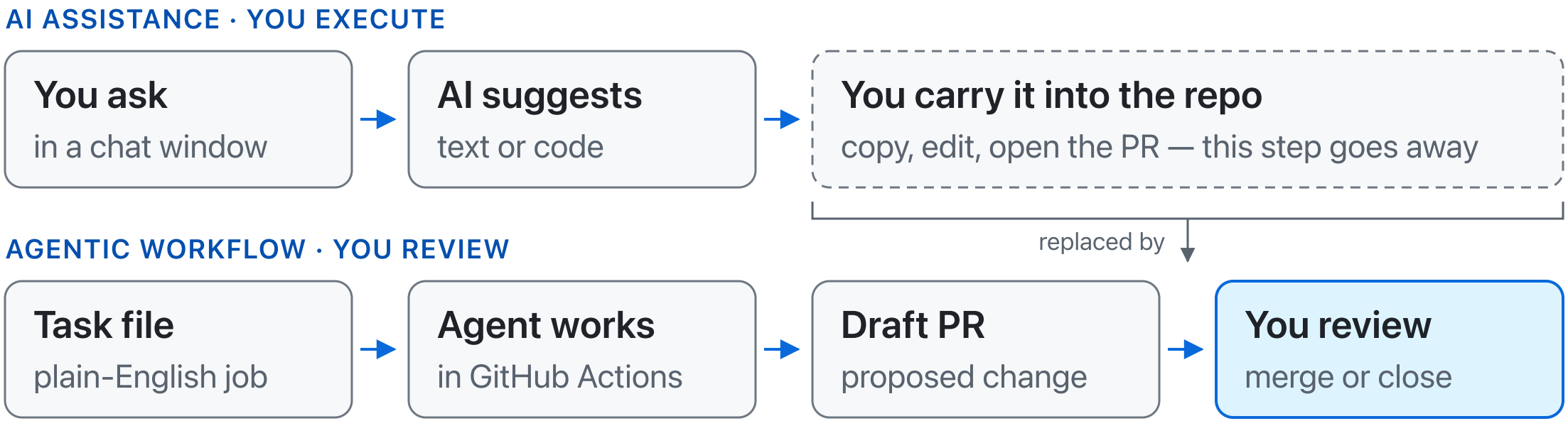
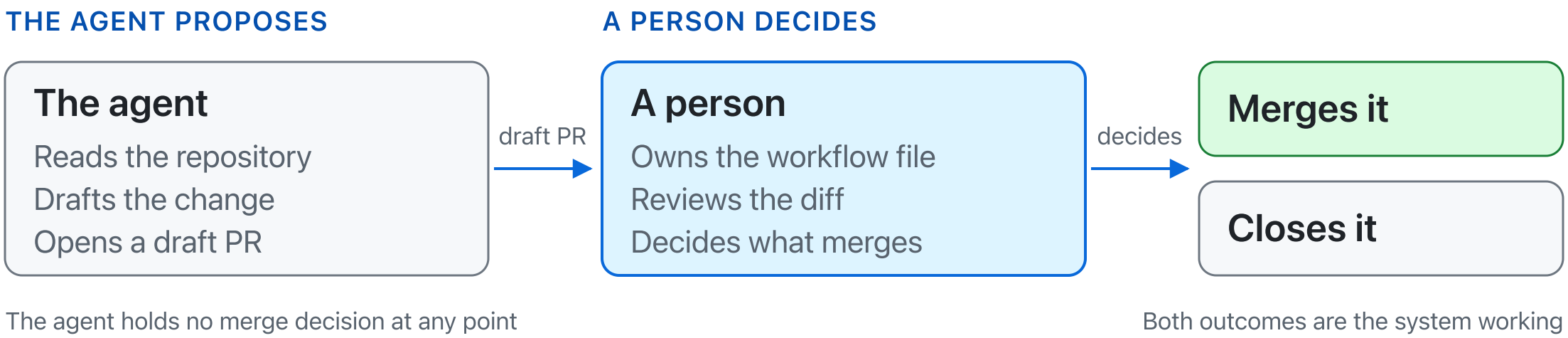


Fig. 2 · From author to editor · illustrative comparison, this deck's framing.

The agent drafts; a person decides what merges



> This moves the work from writing to judging; it does not remove it.

Fig. 3 · This deck's framing · `draft: true` is enforced pull-request creation policy · reviewer minutes are not yet measured.

Only judgement can tell whether the words still match the code

A script, when the rule is fixed

Broken links. Missing API entries. Formatting.

Cheaper, faster, and it cannot surprise you.

An agent, when it needs reading

Does this paragraph still describe what the function does?

No rule expresses that.

> This does not replace the build, test and release pipeline.

This deck's framing · task-fit detail is in the appendix · use the deterministic tool whenever one solves the task.

07 / THE REAL EXAMPLE

The unit of review is one small diff

PUBLIC EXAMPLE · github/gh-aw #8716 · merged 3 Jan 2026 · different workflow, shows output size only

A dialog title was appearing in the page's navigation menu.

BEFORE · UNWANTED TOC ENTRY

```
<h3 id="error-modal-title">
  Error Copying File
</h3>
```

AFTER · ACCESSIBLE MODAL LABEL

```
<div role="heading" aria-level="3"
     class="error-modal-title">
  Error Copying File
</div>
```

> Three lines changed. A reviewer can judge that in seconds.

Diff simplified · Copilot follow-up from a docs-testing chain, not our pilot.

Real PR and diff #8716 · GitHub documentation report

08 / THE EVIDENCE

GitHub reports 57 of 59 documentation pull requests merged

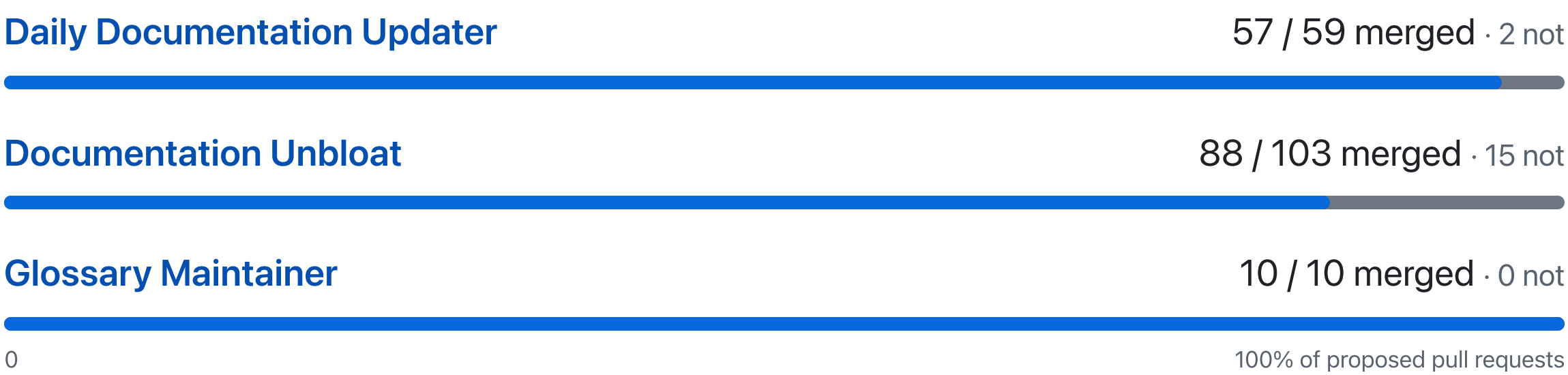


Fig. 4 · GitHub report · 13 Jan 2026 · selected small samples, not a forecast for our pilot.

09 / THE RISK

It will be wrong sometimes, and that has to be survivable

Confidently wrong

It matches a page to code it misread.

Too wide a diff

A one-page fix touches twenty files.

Poisoned input

Instructions hidden in an issue it reads.

Spend, no merge

Credits burn whether or not anything lands.

> Trusting the model prevents none of these; each needs a limit we set.

Failure modes this deck names · prompt injection is documented by GitHub; the rest follow from the task.

TODO · our own error rate is unmeasured — week one's staged preview is the first measurement.

The limits live in a file you can read, not in the model

```
---
on:
  push:
    branches: [main]
permissions: read-all
safe-outputs:
  create-pull-request:
    draft: true
---
# Update Docs
Ensure every code-level change is
mirrored by clear, accurate, and
stylistically consistent documentation.
```

Shift · `on:`
Every push to main

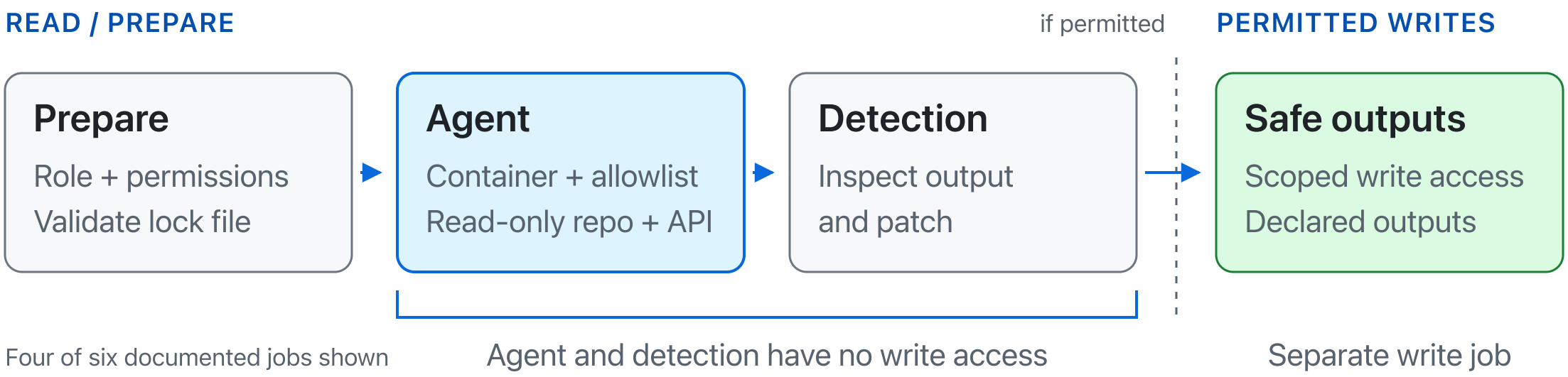
Access badge · `permissions`
Read-only

Outbox · `safe-outputs`
Draft pull requests only

Job description · Markdown body
Plain-English instructions

Real file, trimmed · githubnext/agentics workflows/update-docs.md · PR policy

Only a separate job can write; the agent cannot



> Detection runs before the write job, but by default it cautions rather than blocks.

Fig. 5 · Default `continue-on-error: true` · layers of control, not a guarantee · least privilege (Saltzer & Schroeder, 1975).

A cheap run nobody merges is still expensive

$$C_{\text{acc}} = \frac{p_a A + p_m M + wR}{N_{\text{acc}}}$$

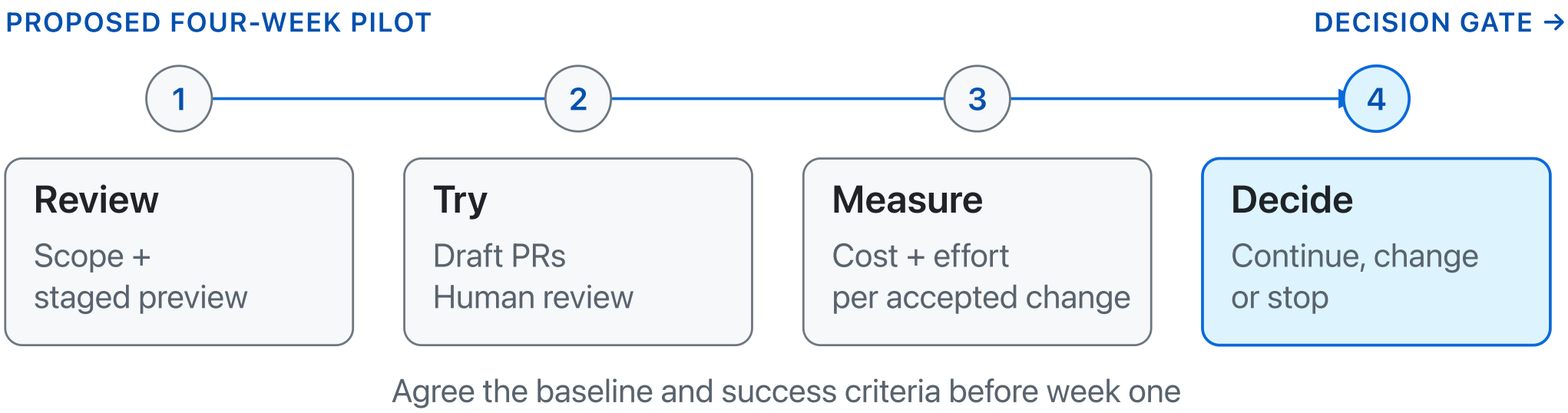
$$C_{\text{man}} = \frac{wR_{\text{man}}}{N_{\text{man}}} \qquad \text{continue if } C_{\text{acc}} < C_{\text{man}}$$

SYMBOL	QUANTITY	STATUS
A	AI credits, AIC	measured, `gh aw logs`
p_a	price per credit	≈0.01, best-effort
M	Actions minutes	measured, billing
p_m	price per minute	TODO
R	reviewer minutes	not measured
w	reviewer cost/min	TODO
N_{acc}	proposals merged	measured

Eq. 1 · 1 AIC ≈ $0.01 estimated · default cap 1,000 AIC per run (≈ $10) · billing may differ.

TODO · Agree pilot limits; measure cost per run and review time.

Four weeks turns opinion into evidence



> Review accepted changes, reviewer effort and total cost together.

Fig. 6 · Week 1 runs with `staged: true`: writes are skipped, the run still costs.

Week four ends in a decision we agreed in advance

Useful fixes accepted

Problems a reviewer agreed were real.

Less total effort

Lower than the agreed manual baseline.

Within budget

Spend inside the caps we set.

> Agreed before we start, so the result can tell us to stop.

TODO · all three thresholds, and the manual baseline they are measured against, need agreeing before week one.

Proposed criteria · this deck's recommendation, not an agreed plan.

The downside is capped before anything runs

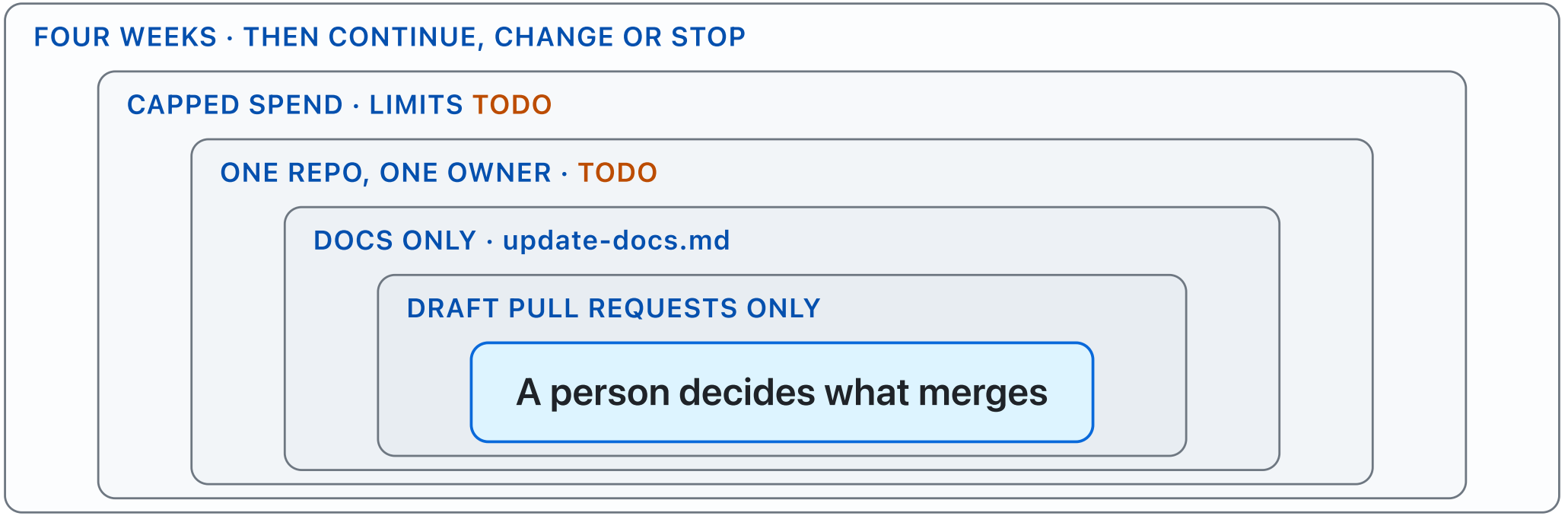


Fig. 7 · Proposed bounds · spend, repository and owner are TODO.

Approve a four-week, docs-only pilot

PROPOSED STARTING POINT

Candidate: `githubnext/agentics workflows/update-docs.md`

Bounds as on the previous slide.
Switch off at any time.

Pilot repository + owner
TODO: name both

Per-run + total pilot budget
TODO: agree explicit limits

Week-four success thresholds
TODO: set before starting

A two-way door: reversible by design · pause on unsafe output or a breached limit.

APPENDIX 1 / COMPILATION

Compilation turns that file into an Actions workflow

1 · Author

Markdown task
and frontmatter

2 · Compile

`gh aw compile`
Generated `.lock.yml`

3 · Publish

Commit both files
Run in GitHub Actions

> Compilation validates configuration and pins Actions to commit SHAs.

Agents add a third lane beside CI and CD

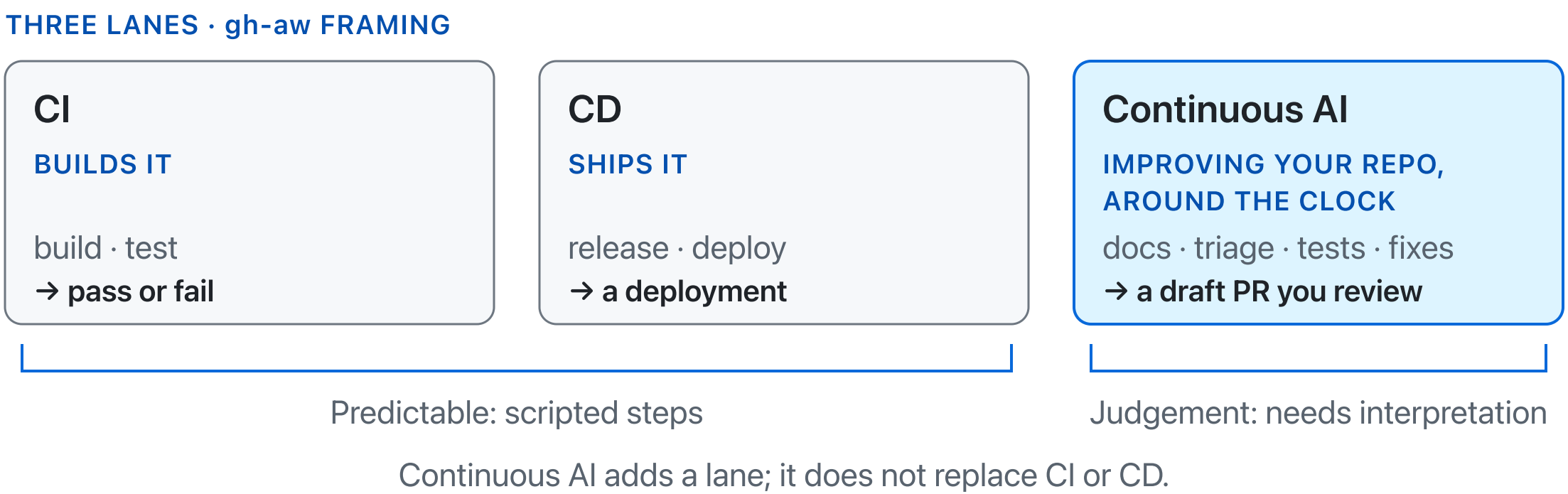


Fig. 8 · Lane captions quoted from the gh-aw docs · the predictable/judgement split is this deck's framing.

gh-aw docs: "A third pillar beside CI and CD"

Detection adds a review gate, not a safety guarantee

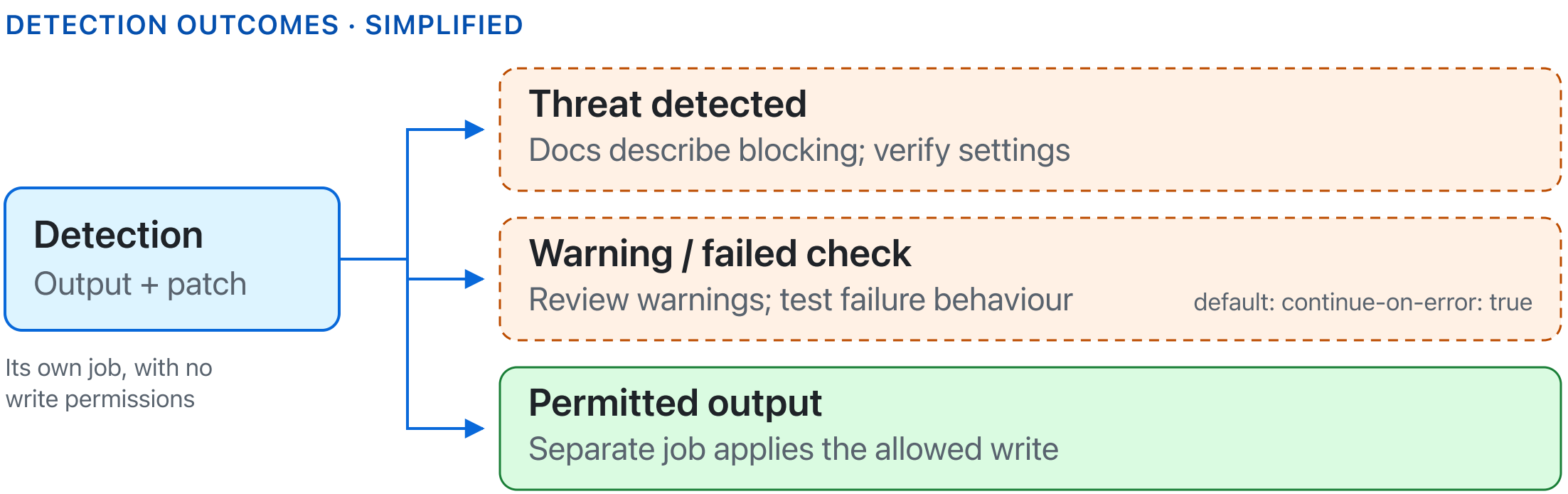


Fig. 9 · Failure behaviour is disputed in the docs; test it with explicit settings before relying on it.

Use agents where interpretation adds value

Good pilot candidate

Compare code with documentation.

Propose a small, reviewable correction.

Keep predictable work in CI

Build, test and release steps.

Use deterministic tools when they solve the task.

TODO · Agree which tasks stay with a person and our current autonomy level.

Start with one workflow from the example collection

Example collection

TODO: repository URL
TODO: workflow file
TODO: recommended first task

Review before adding

Instructions and scope
Permissions and network
Outputs and budget

```
gh aw add-wizard owner/repo/workflow
```

Command template · replace it with the confirmed workflow reference.

A trial can run in a temporary private repository

Select

Choose the reviewed workflow file.

Trial

`gh aw trial <workflow>`
Temporary private repo by default.

Inspect

Read the results before choosing the next step.

> TrialOps is experimental.

Isolation changes where the workflow runs; it does not make the run free or risk-free.

Staged mode previews safe outputs before they are applied

```
safe-outputs:
  staged: true
  create-pull-request:
    draft: true
```

Compile and publish

Commit the Markdown and lock file.

Run and inspect

Read the Actions step summary.

Then decide

Enable draft PRs only after review.

> Staged safe-output writes are skipped; the workflow still runs.

APPENDIX 8 / DETAILS FOR QUESTIONS

The technical claims are grounded in GitHub's documentation

Public preview since 11 Jun 2026

Product and controls

Official changelog
Architecture and threat detection
PR policy and staged mode

Operating and measuring

CLI and schedule syntax
Cost management
Reported documentation results

Unresolved conflicts · daily AI-credit cap default · whether a failed detection run blocks safe outputs

Technical fact check: 17 Oct 2026. Sources are linked on the relevant slides and in speaker notes.

Public-preview changelog · Official docs